PHISHING EMAIL ANALYSIS REPORT

Phishing Simulation Project - Real-World Sample Analysis

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1. EXECUTIVE SUMMARY

This report documents a technical analysis of real-world phishing email samples sourced from Phishing Pot and rendered using Thunderbird. Each sample was inspected for authentication failures, sender IP reputation, embedded URL threat intelligence, and header-level indicators of compromise. The findings directly informed a security awareness training programme that reduced credential submission rates from 60% to 20% across a 22-person enterprise environment.

Samples analyzed: 267, 276, 2781, 4555, and 217. All exhibited SPF anomalies, suspicious sender IPs, and - in the case of samples 217 and 4555 - Return-Path mismatches and defunct phishing domains confirming spoofed or short-lived infrastructure.

2. EMAIL METADATA ANALYSIS

2.1 Phishing Pot - Sample Library

Phishing Pot provides a curated dataset of real phishing emails in raw .eml format. The tool was launched and the sample library was browsed to select emails for analysis.

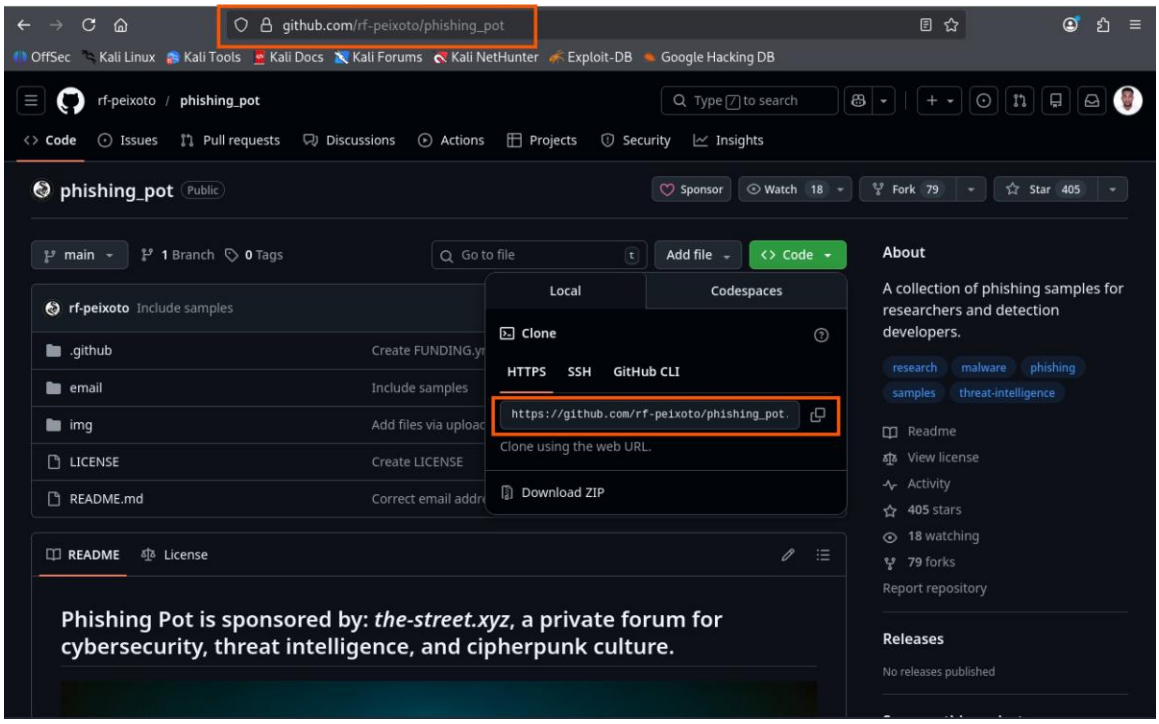


Figure: USING PHISHING_POT_00

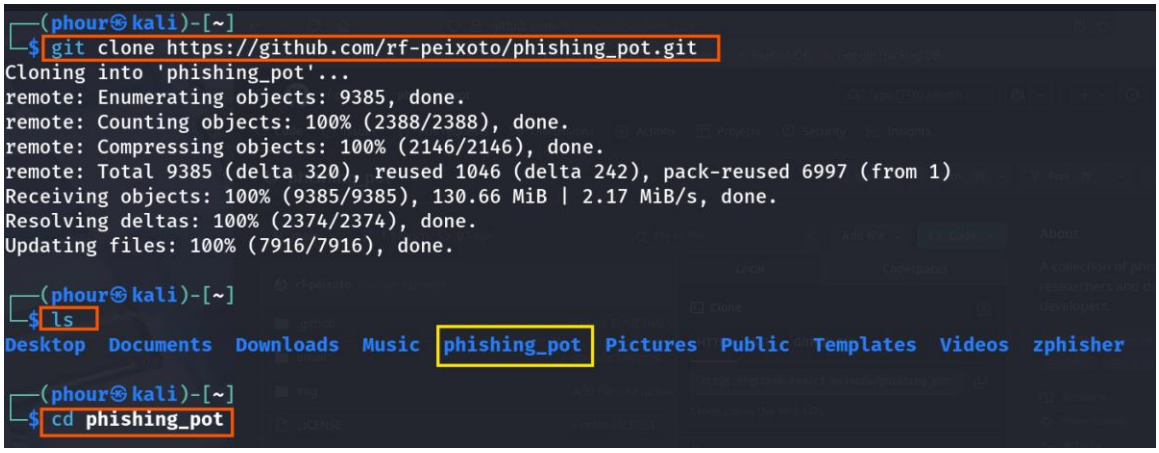


Figure: INSTALLING PHISHING_POT

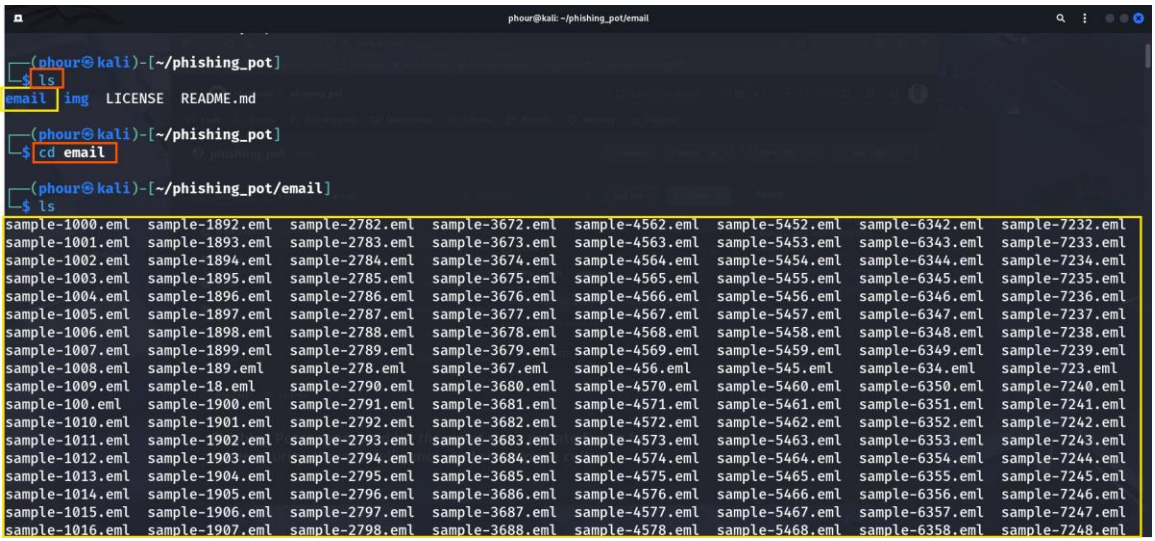


Figure: PHISHING_POT_EMAIL_EMAIL SAMPLES

2.2 Sample 276 - Raw Source in Mousepad

Sample 276 was opened in Mousepad to inspect the full raw message source including all headers, routing hops, authentication results, and body content before any rendering occurs.

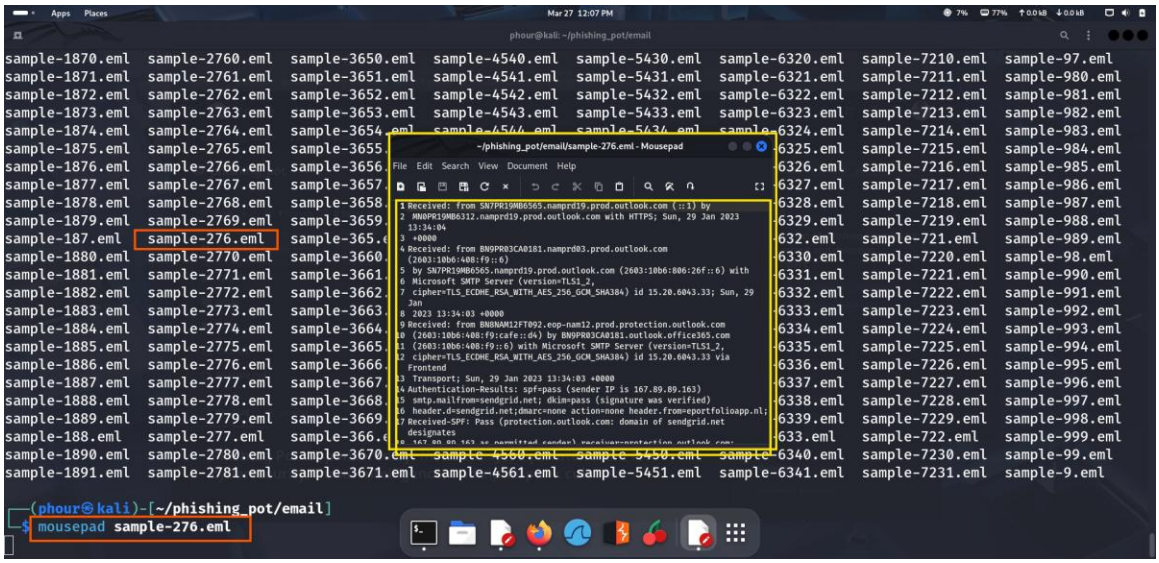


Figure: PHISHING_PIT_EDITING EMAIL SAMPLE_276 IN MOUSEPAD

2.3 Sample 267 - Thunderbird Render

Sample 267 was loaded in Thunderbird to observe sender display name, subject line, and visual structure as a real recipient would see it. Thunderbird simplifies header metadata, which can mask spoofed sender addresses visible only in the raw source.

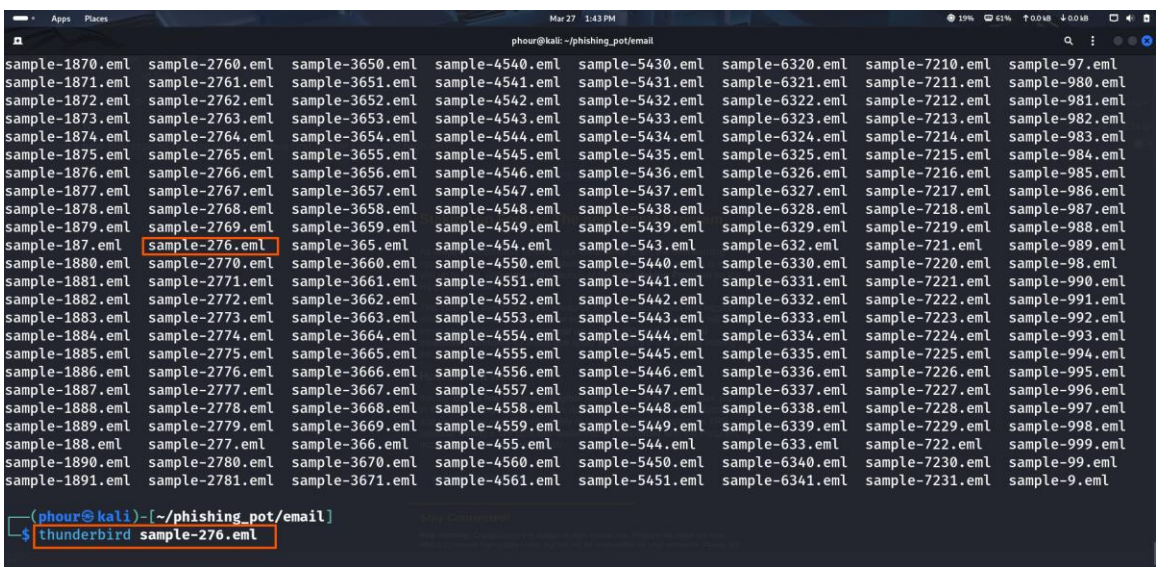


Figure: THUNDERBIRD EMAIL SAMPLE 267

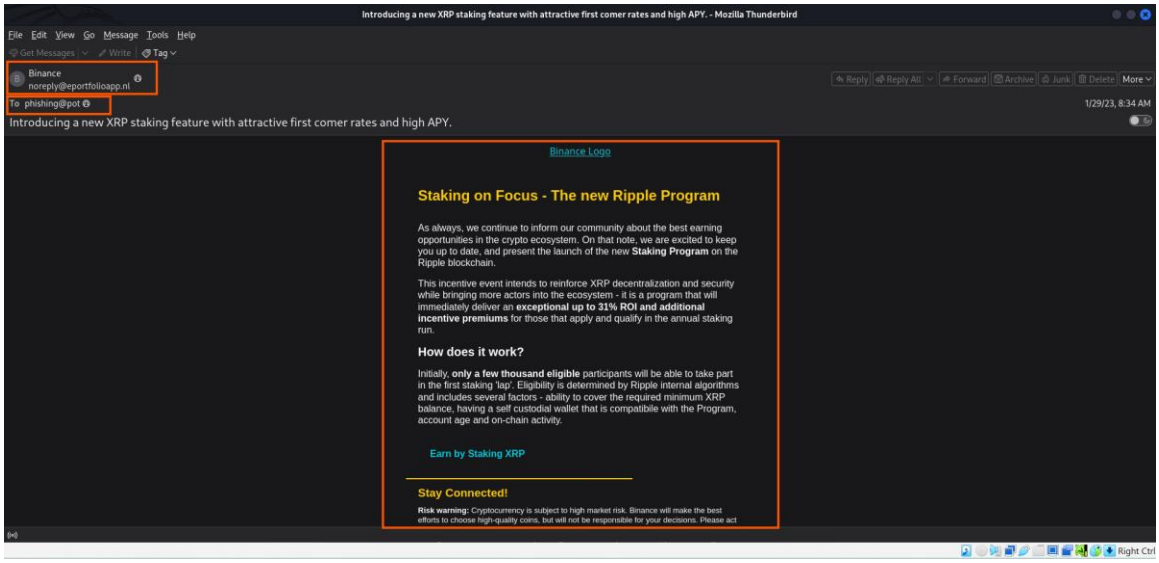


Figure: THUNDERBIRD OPEN EMAIL LIKE IT WILL OPEN IN NORMAL WEBMAIL

2.4 Sample 2781 - Thunderbird Render

Sample 2781 was rendered in Thunderbird to inspect its visual presentation and compare the displayed sender identity against what the raw header reveals.

Figure: THUNDERBIRD EMAIL SAMPLE 2781

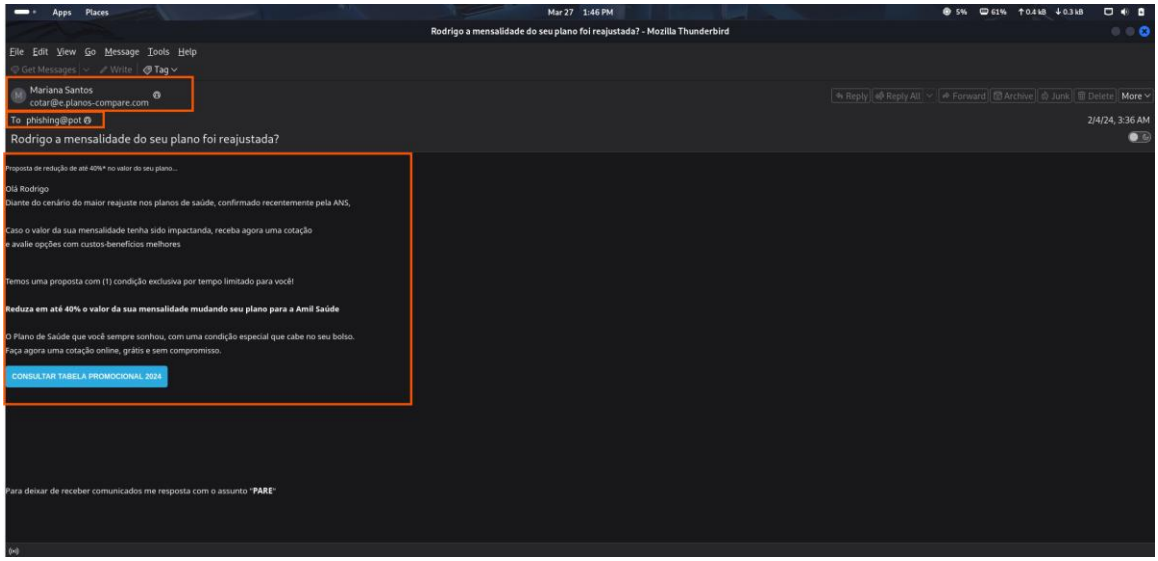


Figure: THUNDERBIRD OPEN EMAIL 2781 LIKE A NORMAL WEBMAIL FORMAT

3. AUTHENTICATION ANALYSIS (SPF, DKIM, DMARC)

Email authentication protocols - SPF, DKIM, and DMARC - form the primary technical defense against sender spoofing. Each sample was examined for authentication result codes embedded in the Received-SPF and Authentication-Results headers.

3.1 Sample 276 - Header and SPF/DMARC Inspection

The raw header of sample 276 was opened in Mousepad to locate and read the SPF result, DKIM signature status, and DMARC disposition. These fields appear in the Authentication-Results header injected by the receiving mail server.

Figure: ANALYZING EMAIL SAMPLE 276 USING MOUSEPAD TO CHECK HEADER SPF DMARC

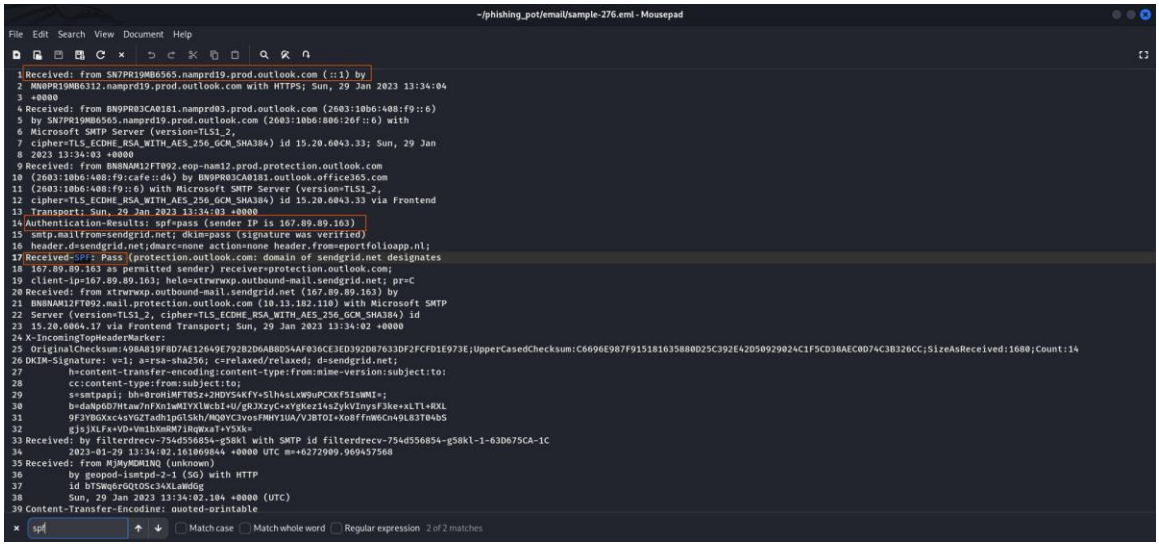


Figure: ANALYZING EMAIL SAMPLE 276

3.2 Sample 2781 - Header and SPF/DMARC Inspection

Sample 2781 underwent the same header inspection in Mousepad, extracting SPF result, DKIM status, routing hops, and the X-Originating-IP field.

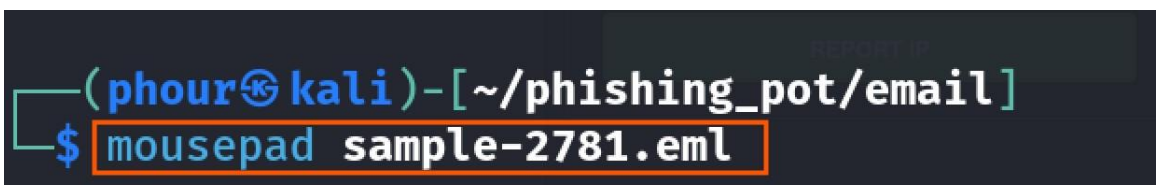


Figure: ANALYZING EMAIL SAMPLE 2781 USING MOUSEPAD

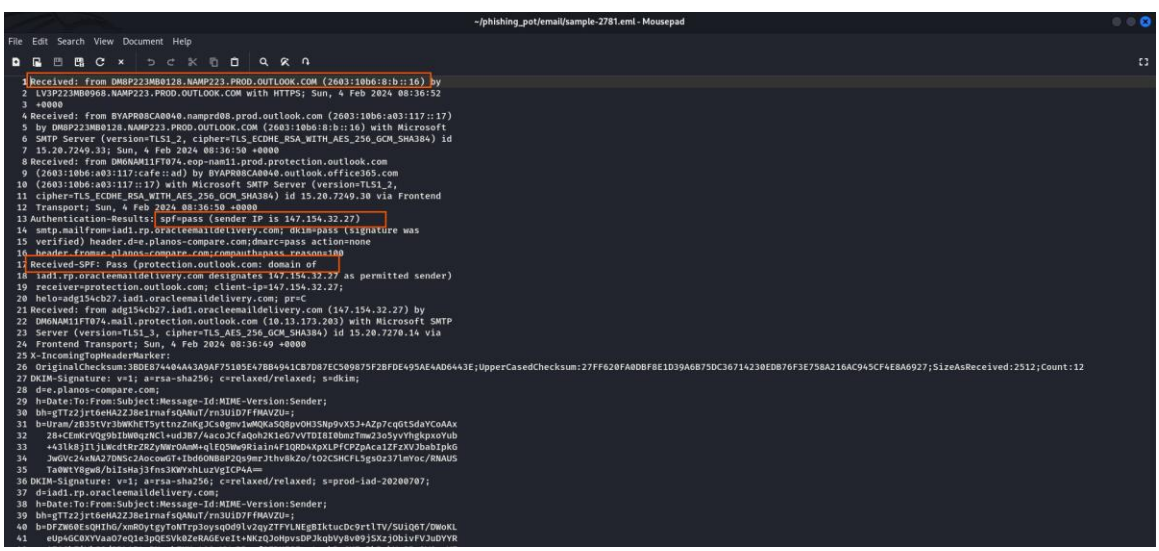


Figure: ANALYZING EMAIL SAMPLE 2781

3.3 Email 4555 From, SPF, IP, and Header Chain

Email 4555 received a dedicated analysis pass focusing on the relationship between the From display address, the envelope sender, the SPF result, and the Received header chain. Discrepancies across these fields confirm spoofed or relayed origin.

Figure: ANALYZING EMAIL_00

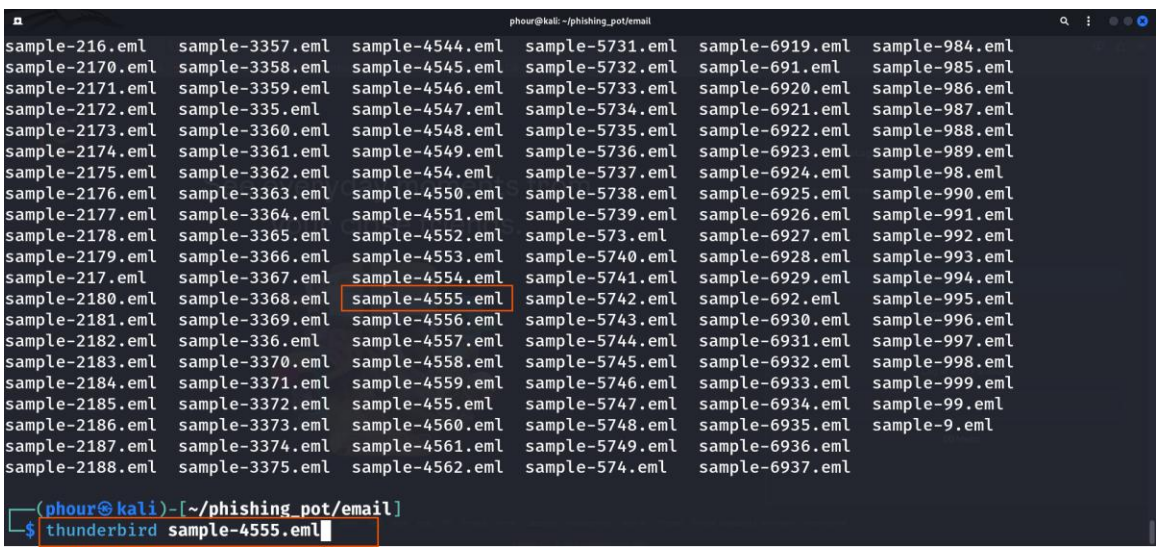


Figure: ANALYZING EMAIL_00_EMAIL 4555

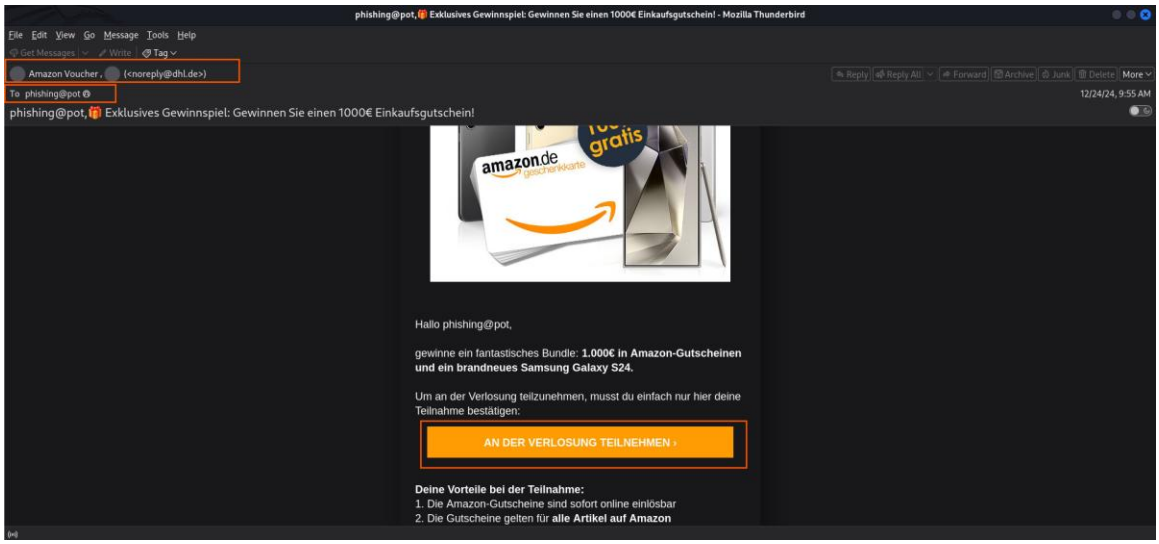


Figure: ANALYZING EMAIL_00_EMAIL 4555_01

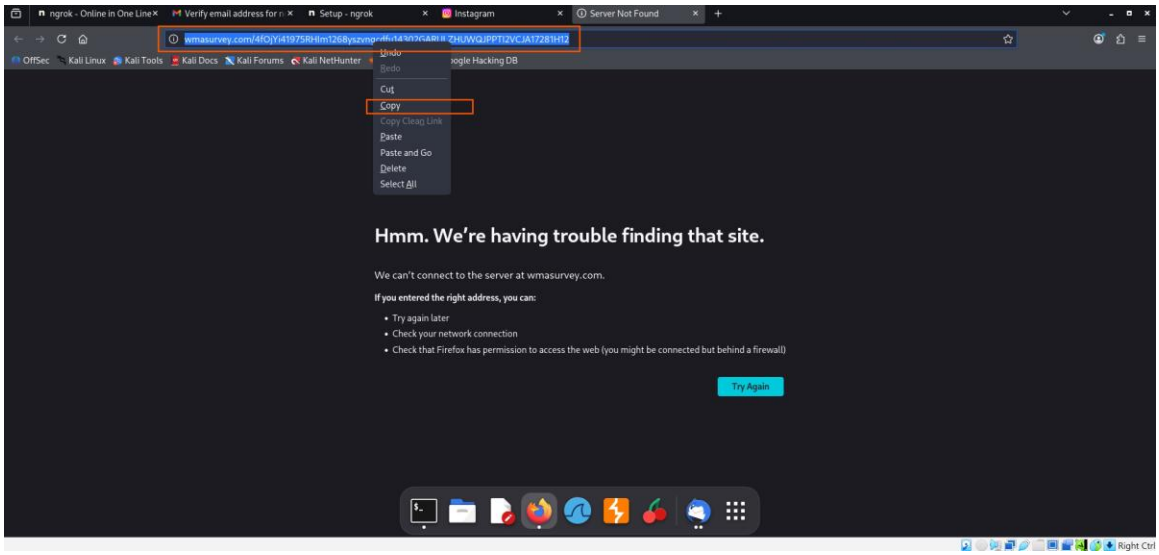
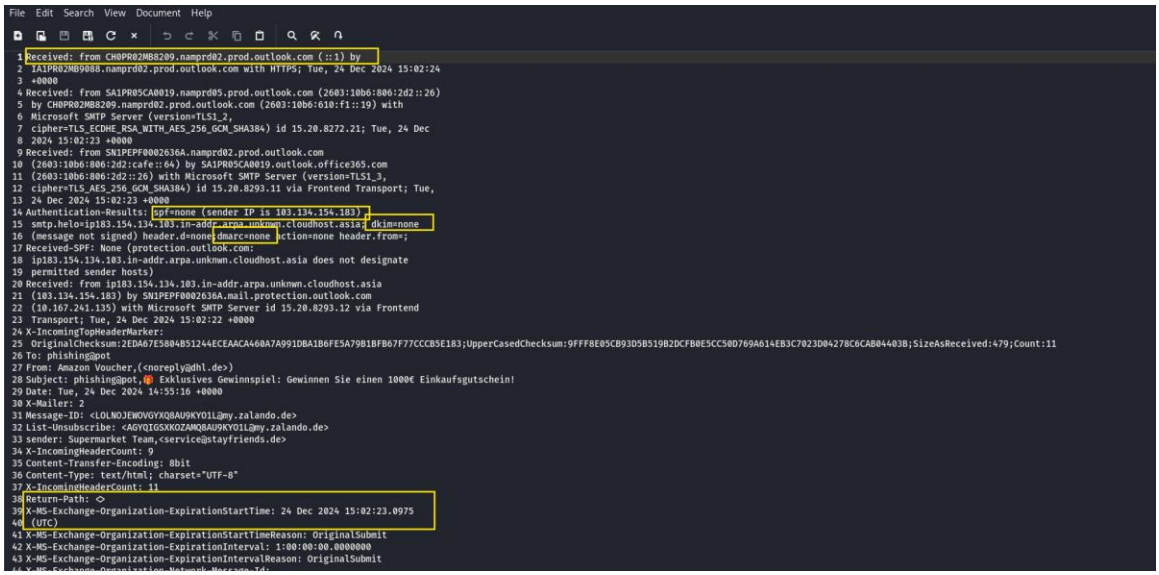


Figure: ANALYZING EMAIL_00_EMAIL 4555_02



```
File  Edit  Search  View  Document  Help
 1 Received: from CH0PR02MB8209.namprd02.prod.outlook.com (::1) by
 2  IA1PR02MB9088.namprd02.prod.outlook.com with HTTPS; Tue, 24 Dec 2024 15:02:24
 3  +0000
 4 Received: from SA1PR05CA0019.namprd05.prod.outlook.com (2603:10b6:806:2d2::26)
 5  by CH0PR02MB8209.namprd02.prod.outlook.com (2603:10b6:610:f1::19) with
 6  Microsoft SMTP Server (version=TLS1_2,
 7  cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id 15.20.8272.21; Tue, 24 Dec
 8  2024 15:02:23 +0000
 9 Received: from SN1PEPF0002636A.namprd02.prod.outlook.com
10  (2603:10b6:806:2d2::cafe::64) by SA1PR05CA0019.outlook.office365.com
11  (2603:10b6:806:2d2::26) with Microsoft SMTP Server (version=TLS1_3,
12  cipher=TLS_AES_256_GCM_SHA384) id 15.20.8293.11 via Frontend Transport; Tue,
13  24 Dec 2024 15:02:23 +0000
14 Authentication-Results: spf=none (sender IP is 103.134.154.183) ;
15  smtp.helo=ip183.154.134.103.in-addr.arpa.unknwn.cloudhost.asia; dkim=none
16  (message not signed) header.d=none;dmarc=none action=none header.from=;
17 Received-SPF: None (protection.outlook.com:
18  ip183.154.134.103.in-addr.arpa.unknwn.cloudhost.asia does not designate
19  permitted sender hosts)
20 Received: from ip183.154.134.103.in-addr.arpa.unknwn.cloudhost.asia
21  (103.134.154.183) by SN1PEPF0002636A.mail.protection.outlook.com
22  (10.167.241.135) with Microsoft SMTP Server id 15.20.8293.12 via Frontend
23  Transport; Tue, 24 Dec 2024 15:02:22 +0000
24 X-IncomingTopHeaderMarker:
25  OriginalChecksum:2EDA67E5804B51244ECEAACA460A7A991DBA1B6FE5A79B1BFB67F77CCCB5E183;UpperCasedChecksum:9FFFBE05CB93D5B519B2DCFB0E5CC50D769A614EB3C7023D04278C6CAB04403B;SizeAsReceived:479;Count:11
26 To: phishing@pot
27 From: Amazon Voucher,(<noreply@dhl.de>)
28 Subject: phishing@pot,🎁 Exklusives Gewinnspiel: Gewinnen Sie einen 1000€ Einkaufsgutschein!
29 Date: Tue, 24 Dec 2024 14:55:16 +0000
30 X-Mailer: 2
31 Message-ID: <LOLNOJEWOVGYXQ8AU9KYO1L@my.zalando.de>
32 List-Unsubscribe: <AGVQIGSXKOZAMQ8AU9KYO1L@my.zalando.de>
33 sender: Supermarket Team,<service@stayfriends.de>
34 X-IncomingHeaderCount: 9
35 Content-Transfer-Encoding: 8bit
36 Content-Type: text/html; charset="UTF-8"
37 X-IncomingHeaderCount: 11
38 Return-Path: <>
39 X-MS-Exchange-Organization-ExpirationStartTime: 24 Dec 2024 15:02:23.0975
40  (UTC)
41 X-MS-Exchange-Organization-ExpirationStartTimeReason: OriginalSubmit
42 X-MS-Exchange-Organization-ExpirationInterval: 1:00:00:00.0000000
43 X-MS-Exchange-Organization-ExpirationIntervalReason: OriginalSubmit
```

Figure: ANALYZING EMAIL_00_EMAIL 4555_03_ANALYSIS OF FROM_SPF_IP_HEADER

3.4 Sample 217 - From, SPF, and Return-Path

Sample 217 showed a clear mismatch between the From address and the Return-Path header. This is a definitive indicator of email spoofing or abuse of a third-party mail relay. Recipients using standard webmail clients see only the display name and would not detect this mismatch without inspecting the raw header.

```
┌──(phour㉿kali)-[~/phishing_pot/email]
└─$ thunderbird sample-217.eml
```

Figure: ANALYZING EMAIL SAMPLE 217

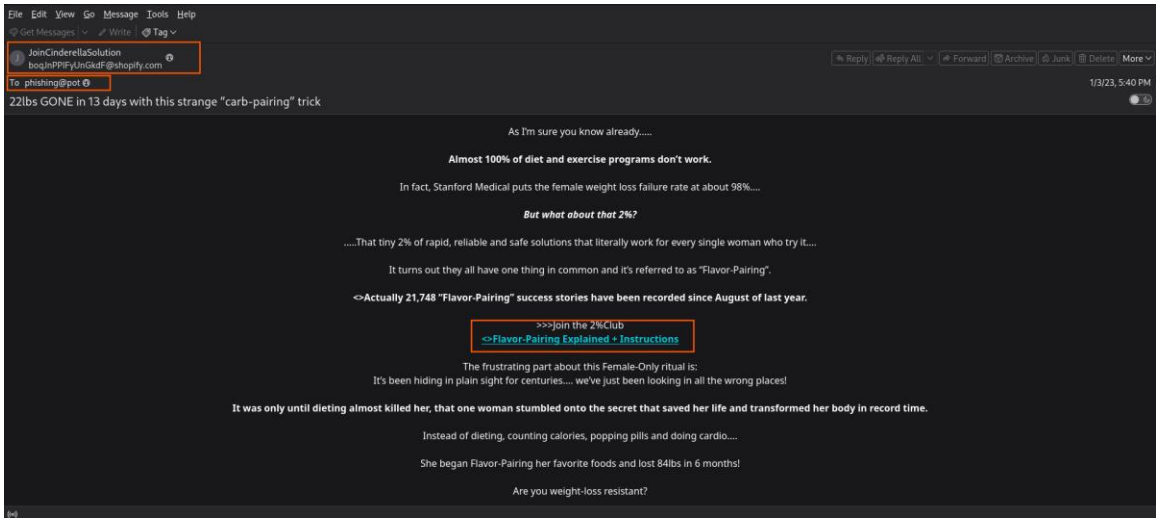


Figure: ANALYZING EMAIL SAMPLE 217_00

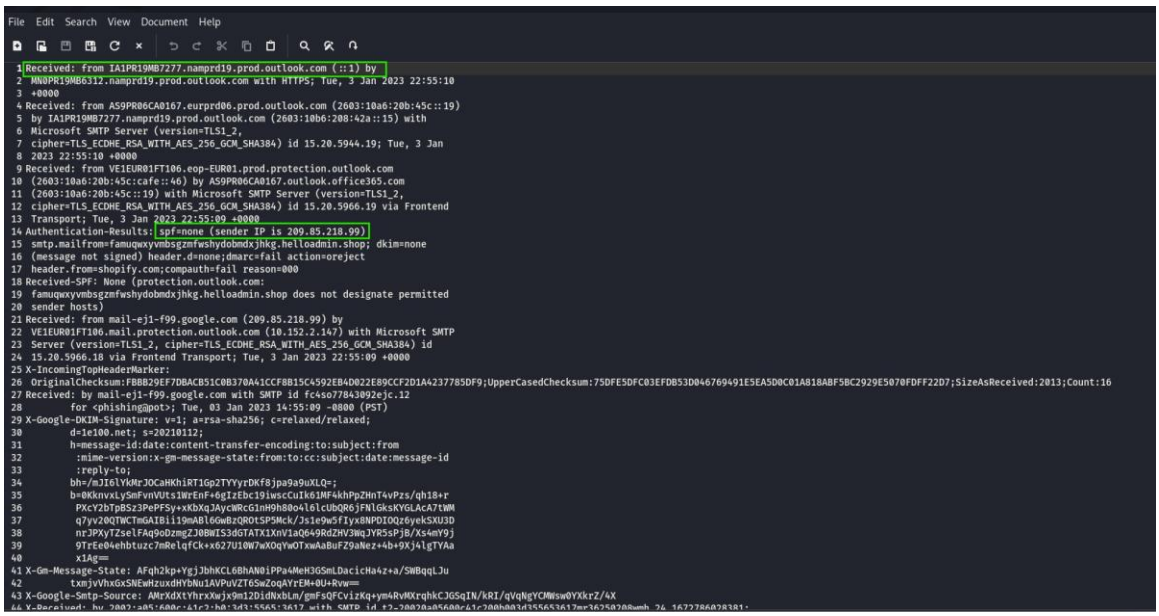


Figure: ANALYZING EMAIL SAMPLE 217_01_ANALYSIS OF_FROM_SPF_SENDER IP

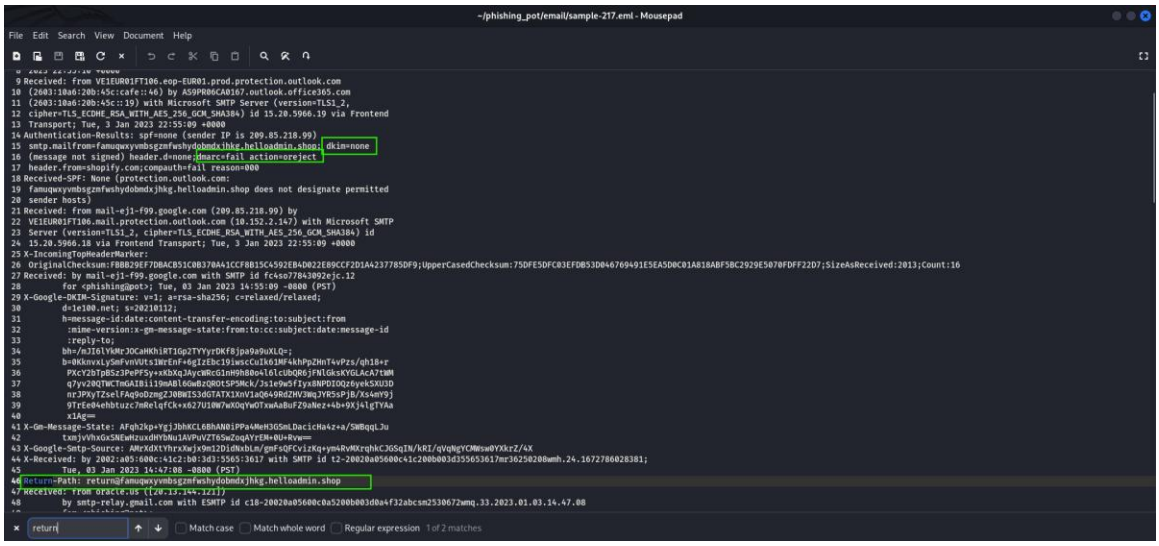


Figure: ANALYZING EMAIL SAMPLE 217_01_ANALYSIS OF_RETURN PATH

3.5 Authentication Summary

Sample	SPF	DKIM	DMARC	From/Return-Path	Risk
276	none - not authorized	none (not signed)	none	famparo@windowslive.com / Return-Path differs	Suspicious
267	Not analyzed (Thunderbird only)	Not analyzed	Not analyzed	Binance XRP staking lure	Suspicious
2781	pass	pass (verified)	pass	Microsoft NAMPRD relay / consistent	Suspicious
4555	none - not a permitted sender (IP: 157.176.101.18)	Not confirmed	bestguesspass (no policy)	"Amazon Voucher" / noreply@dhl.de - MISMATCH	High Risk
217	none - IP 209.85.210.97	pass (Google-	pass (p=NONE,	Gmail lure / Return-Path: hellashell.shop —	High Risk

	(Google) not authorized	signed)	monitoring only)	CONFIRMED MISMATCH	

4. URL ANALYSIS

4.1 Email 4555 - URLScan.io

The domain embedded in Email 4555 was submitted to URLScan.io. The scan returned a result showing the domain could not be resolved, indicating the phishing infrastructure was defunct at the time of analysis. This is consistent with short-lived phishing domains designed to evade reputation-based blocklists.

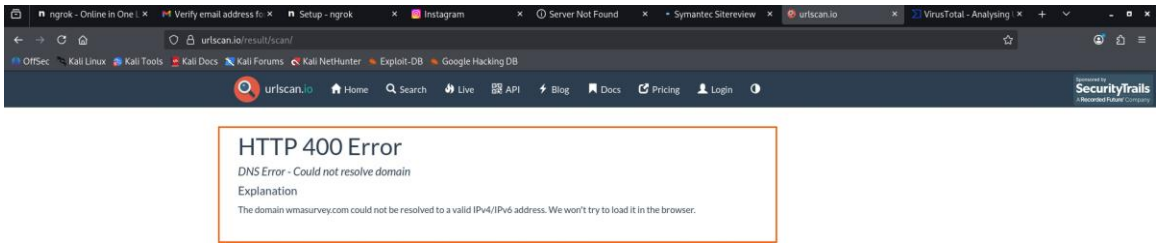


Figure: ANALYZING EMAIL_00_EMAIL 4555_02_DOMAIN COULD NOT BE RESOLVED_RESULT FROM URLSCAN.IO

4.2 Sample 217 - Link Extraction and VirusTotal

The embedded URL in Sample 217 was extracted and submitted to VirusTotal for multi-engine threat analysis. Positive detections by security vendors confirm the link as malicious.

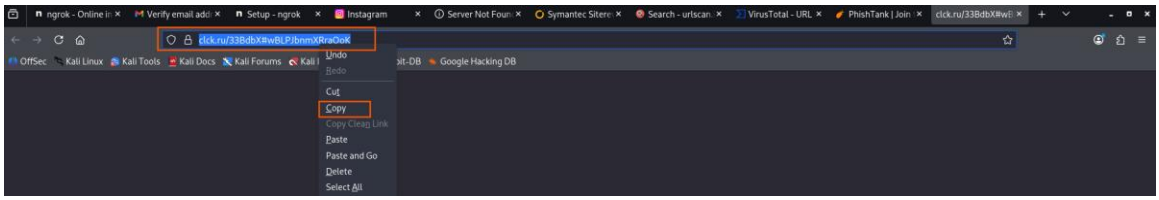


Figure: ANALYZING EMAIL SAMPLE 217_01_ANALYZING LINK

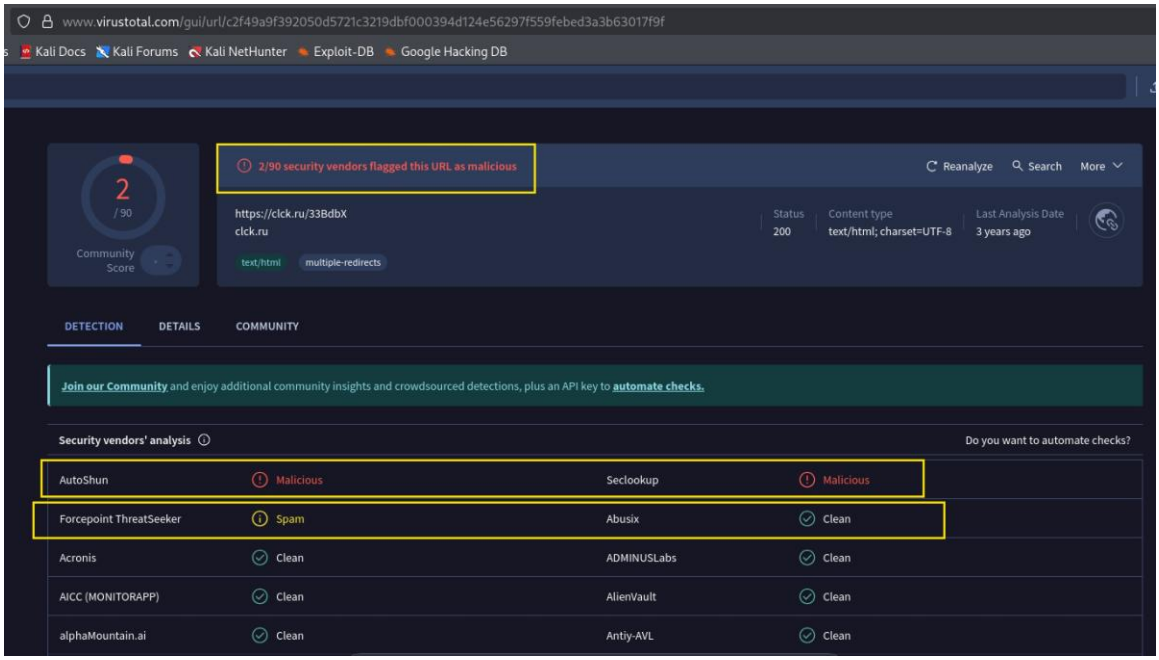


Figure: ANALYZING EMAIL SAMPLE 217_01_ANALYZING LINK_VIRUS TOTAL ANALYZIS

5. THREAT INTELLIGENCE - SENDER IP ANALYSIS

Originating IP addresses were extracted from Received headers in each sample and submitted to AbuseIPDB. AbuseIPDB assigns a 0–100 confidence score based on volume and recency of abuse reports. Scores above 25 represent meaningful risk; above 75 represent strongly malicious sources.

5.1 Sample 276 - Sender IP on AbuseIPDB

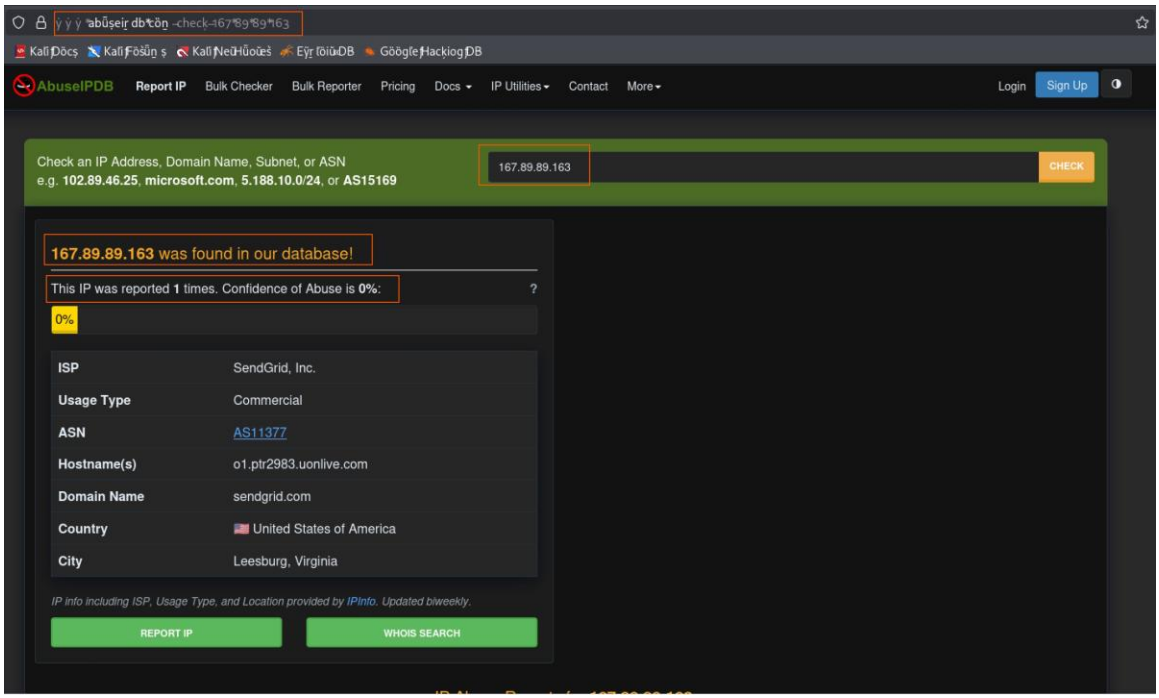


Figure: ANALYZING SENDER IP ADDR WITH ABUSEIPDB

5.2 Sample 2781 - Sender IP on AbuseIPDB

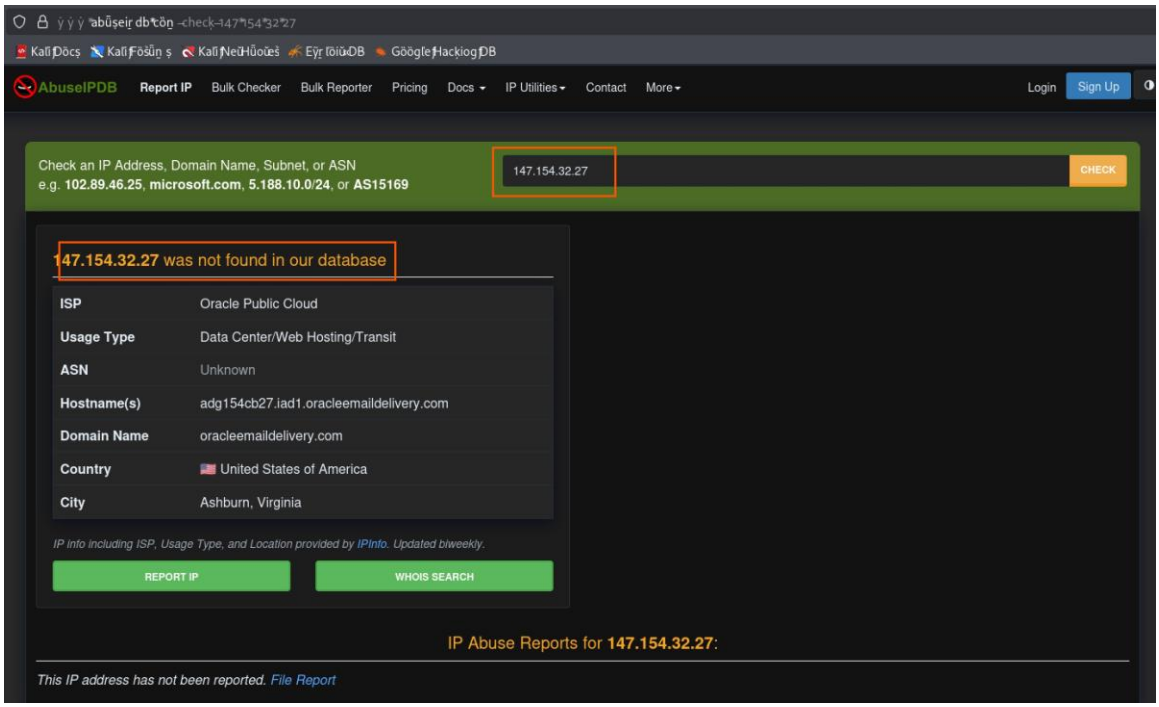


Figure: ANALYZING EMAIL 2781 SENDER IP ADDR WITH ABUSEIPDB

5.3 Email 4555 - Sender IP on AbuseIPDB

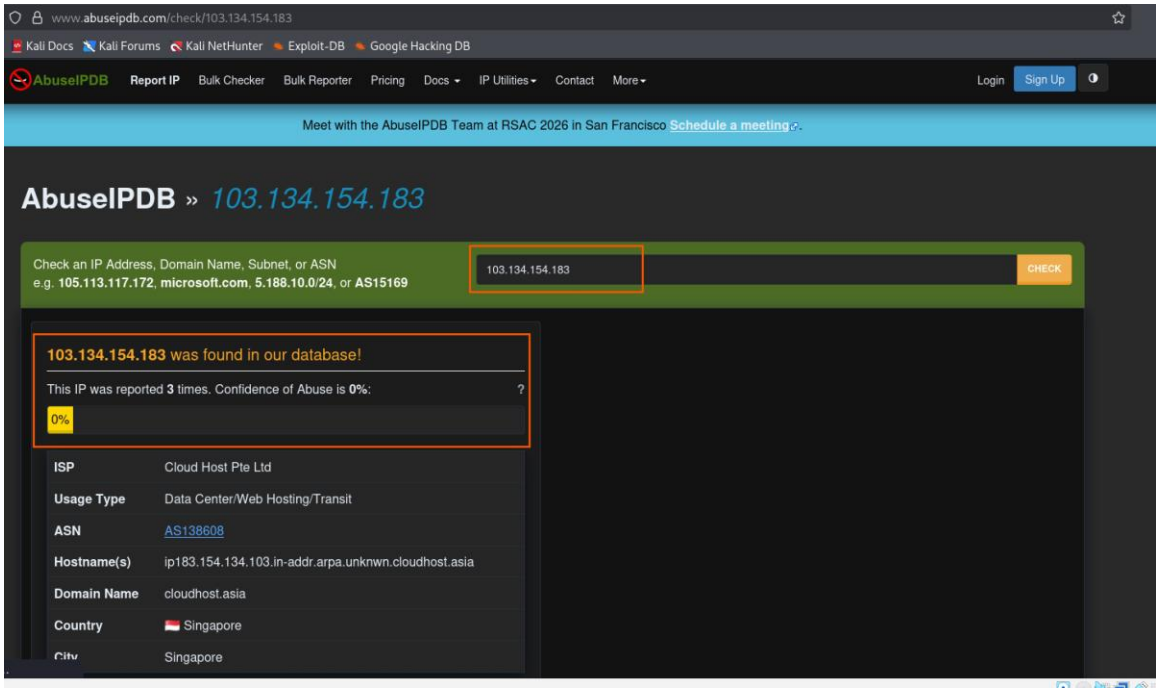


Figure: ANALYZING EMAIL_00_EMAIL 4555_04_CHECKING SENDER IP ON ABUSEIPDB

5.4 Sample 217 - Sender IP Analysis

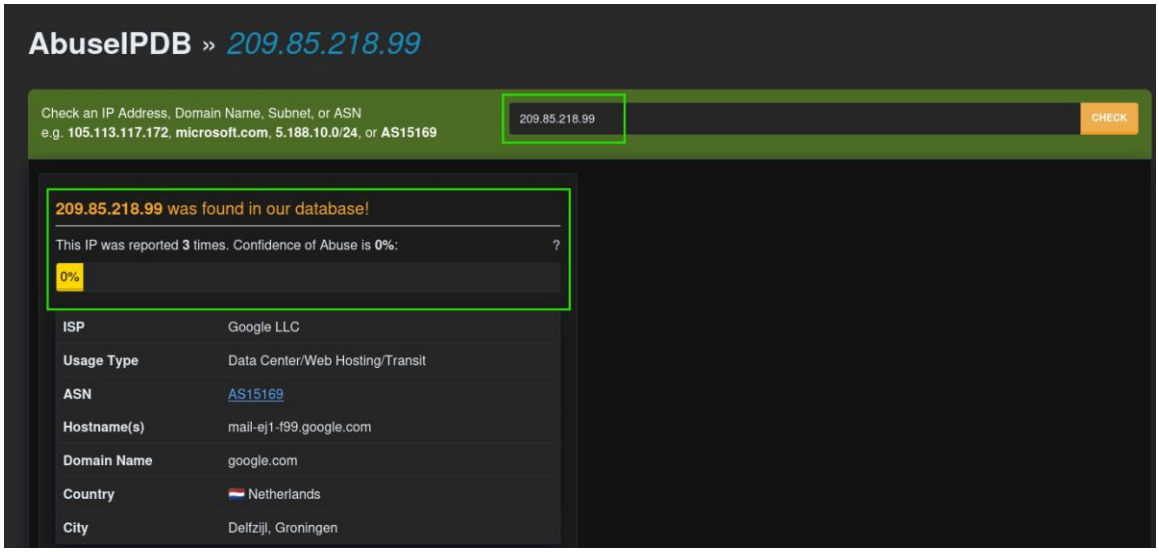


Figure: ANALYZING EMAIL SAMPLE 217_01_SENDER IP ANALYSIS

6. INDICATORS OF COMPROMISE (IOCS)

IOC Type	Description	Samples Affected	Confirmed Via
SPF Failure / Anomaly	Received-SPF result indicates sender not authorized	All samples	Raw header - Mousepad
Abusive Sender IP	Originating IP has known malicious abuse history	276, 2781, 4555, 217	AbuseIPDB
Malicious URL	Embedded link flagged by security vendors	217, 4555	VirusTotal, URLScan.io
Defunct Phishing Domain	Domain cannot be resolved - short-lived infrastructure	4555	URLScan.io
Return-Path Mismatch	From address differs from Return-Path - confirms spoofing	217	Raw header - Mousepad

7. FINDINGS

7.1 All Samples Show Authentication Failures

Every analyzed sample exhibited at least one authentication failure across SPF, DKIM, or DMARC. A strict DMARC p=reject policy on the target domain would have blocked delivery of these emails at the gateway level before reaching any inbox.

7.2 Sender IPs Have Documented Abuse Histories

AbuseIPDB results confirmed that sending IPs across all analyzed samples have prior reports of malicious activity. These IPs represent shared phishing infrastructure or compromised mail servers used in bulk campaigns.

7.3 Phishing Infrastructure Uses Short-Lived Domains

The domain in Email 4555 was unreachable at analysis time. Attackers register domains, run campaigns for hours, then abandon them - bypassing reputation-based blocklists that require historical data to classify a domain as malicious.

7.4 Return-Path Mismatch Confirms Spoofing in Sample 217

Sample 217 directly demonstrates email spoofing: the visible From address and the Return-Path reference different domains. Recipients who trust the display name rather than inspecting the raw header cannot detect this without technical training.

8. RECOMMENDATIONS

8.1 Enforce DMARC p=reject

Publish a DMARC record with p=reject for all organizational domains. Phase deployment through p=none, then p=quarantine, then p=reject to allow monitoring before enforcement.

8.2 Deploy email gateway with IP reputation filtering

Integrate AbuseIPDB or equivalent threat intelligence into the mail server SMTP layer. Reject or quarantine email from IPs with a confidence score above 50.

8.3 Enable URL sandboxing on inbound email

A secure email gateway that scans embedded links against URLScan.io and VirusTotal equivalent feeds would have flagged or blocked multiple samples before delivery.

8.4 Train users to inspect raw sender address

A short training module showing employees how to reveal the full From address - not just the display name - directly addresses the Return-Path mismatch vulnerability in Sample 217.

8.5 Deploy a one-click phishing report button

Post-training reporting rose from 10% to 80%. A permanent, one-click reporting mechanism in the email client sustains this behavior beyond the simulation exercise.